What is a Set in Python?

A **set** is an **unordered**, **unindexed** collection of **unique** items.
It's used when you want to store values **without duplicates** and perform **mathematical operations** like union, intersection, and difference.

🧠 **Syntax**

set_name = {item1, item2, item3}

Example:

fruits = {"apple", "banana", "cherry"}

print(fruits)

Output (order may vary):

{'banana', 'apple', 'cherry'}

🔹 **Characteristics of Sets**

Property	Description
Unordered	Items have no defined order.
Unindexed	You can't access elements using indexes.
No duplicates	Automatically removes duplicates.
Mutable	You can add or remove items.

🔹 **Example: Duplicates are removed automatically**

numbers = {1, 2, 3, 3, 4, 4, 5}

print(numbers)

Output:

{1, 2, 3, 4, 5}

🔹 **Creating a Set**

my_set = {10, 20, 30}

print(type(my_set)) # <class 'set'>

Empty Set:

empty_set = set() # ✅ correct

empty_dict = {}   # ❌ creates a dictionary, not a set

🔹 **Accessing Set Elements**

You **cannot** use indexing like my_set[0] — it will raise an error.
Instead, you can use a loop:

colors = {"red", "green", "blue"}

for color in colors:

    print(color)

🔹 **Adding Elements**

numbers = {1, 2, 3}

numbers.add(4)

print(numbers)

Adding multiple elements:

numbers.update([5, 6, 7])

print(numbers)

🔹 **Removing Elements**

numbers = {1, 2, 3, 4, 5}

```
numbers.remove(3)  # removes 3, raises error if not found
numbers.discard(10)  # no error if not found
print(numbers)
```

Clear all items:

```
numbers.clear()
print(numbers)  # set()
```

🔹 **Set Operations (Very Important)**

Sets support **mathematical operations** like union, intersection, difference, etc.

1️⃣ **Union (combine items)**

```
a = {1, 2, 3}
b = {3, 4, 5}
print(a.union(b))     # {1, 2, 3, 4, 5}
```

or a | b

2️⃣ **Intersection (common items)**

```
print(a.intersection(b))   # {3}
```

or a & b

3️⃣ **Difference (items in a not in b)**

```
print(a.difference(b))     # {1, 2}
```

or a - b

4️⃣ **Symmetric Difference (items not common)**

```
print(a.symmetric_difference(b))   # {1, 2, 4, 5}
```

or a ^ b

🔹 **Checking Membership**

```
colors = {"red", "green", "blue"}
print("red" in colors)   # True
print("yellow" not in colors)  # True
```

🔹 **Set Comparison Operations**

```
a = {1, 2, 3}
b = {1, 2}

print(a.issubset(b))      # False
print(b.issubset(a))      # True
print(a.issuperset(b))    # True
print(a.isdisjoint({4, 5}))  # True
```

🔹 **Copying a Set**

```
original = {1, 2, 3}
new_set = original.copy()
print(new_set)
```

🔹 **Frozen Sets (Immutable Sets)**

If you want a **read-only set**, use frozenset():

```
fset = frozenset({1, 2, 3})
print(fset)

# fset.add(4) ❌ Error: 'frozenset' object has no attribute 'add'
```

🧠 **Summary**

Operation	Method	Example
Add	.add()	s.add(5)
Remove	.remove() / .discard()	s.remove(3)
Union	.union() / `	`
Intersection	.intersection() / &	a & b
Difference	.difference() / -	a - b
Symmetric Difference	.symmetric_difference() / ^	a ^ b
Subset	.issubset()	a.issubset(b)